

Abstract

Existing environmental governance frameworks are structurally inadequate for generative AI systems hosted on cloud infrastructure. Designed around physical supply chains in which the emitting actor owns or controls the emission source, instruments such as the GHG Protocol's Scope 1/2/3 classification, the EU AI Act's energy provisions, and hyperscaler voluntary disclosure programs systematically fail to account for the full computational lifecycle of foundation models. This paper identifies a scoping failure at the core of current AI sustainability governance: the data acquisition, curation, and preprocessing phase, where training energy cost is first and most consequentially determined, falls entirely outside the boundary of every major existing regulatory instrument.

Drawing on the generative AI and cloud infrastructure ecosystem as its primary case, this paper makes two interconnected arguments. First, the regulatory gap is not a measurement problem amenable to incremental disclosure improvements; it is a structural consequence of applying legacy frameworks to a novel industrial system. The paper maps this gap across five lifecycle phases (data acquisition, model training, fine-tuning and alignment, inference at scale, and hardware manufacturing) and demonstrates how each phase exposes a distinct failure mode in current governance architecture. Second, closing this gap requires reconceptualizing data governance as a form of sustainability governance. Dataset curation decisions regarding size, composition, deduplication strategy, and provenance have direct and quantifiable implications for training energy intensity, yet are made entirely outside any environmental regulatory framework.

To operationalize this reconceptualization, the paper proposes three governance mechanisms: a dataset efficiency standard grounded in the data-centric AI literature; a tiered training data licensing framework modeled on music industry rights management; and a mandatory pre-training environmental impact assessment triggered by compute thresholds, analogous to existing industrial impact assessment requirements. These mechanisms are situated within a broader regulatory architecture built on four design principles: lifecycle completeness, jurisdictional coordination, compute-proportional disclosure, and data-layer integration. The paper proposes hyperscalers as regulated intermediaries for enforcement, drawing on the precedent of GDPR data processor obligations, and addresses the concentration risk the framework entails through its tiered compliance structure. The paper concludes by situating this proposal against the EU AI Act, California SB 1047 and SB 53, and the CORSIA aviation emissions standard, arguing that existing instruments confirm the political viability of the governance logic proposed here while leaving the environmental dimension of the data layer entirely unaddressed.

Keywords: AI sustainability, data governance, generative AI, carbon accounting, GHG Protocol, foundation models, cloud infrastructure, regulatory frameworks, data-centric AI, environmental policy